MIL-S-43961A
13 July 1978
SUPERSEDING
MIL-S-43961(GL)
22 February 1977

MILITARY SPECIFICATION

* SHOES, GYMNASIUM, HEAVY DUTY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for heavy duty athletic shoes.

* 1.2 Classification. The shoes shall be in two widths, narrow and medium and of the following sizes as specified, except that the narrow width is applicable only to sizes 3 through 10 whole and half sizes (see 6.2).

* Sizes - 3, 3-1/2, 4, 4-1/2, 5, 5-1/2, 6, 6-1/2, 7, 7-1/2, 8, 8-1/2, 9, 9-1/2, 10, 10-1/2, 11, 11-1/2, 12, 12-1/2, 13, 13-1/2, 14, 14-1/2, and 15.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-L-61	-	Laces, Nylon.
V-T-285	-	Thread, Polyester.
V-T-295	-	Thread, Nylon.
CCC-C-419	-	Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8430

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MIL-T-43548 - Thread, Polyester, Cotton-Covered.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.
FED-STD-601 - Rubber: Sampling and Testing.
FED-STD-751 - Stitches, Seams, and Stitchings.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1056-73 - Sponge and Expanded Cellular Rubber Products

D 1630-61 - Abrasion Resistance of Rubber Compounds for Soles and Heels

D 1667-76 - Flexible Cellular Materials-Vinyl Cloride Polymers and Copolymers (Closed Cell Sponge)

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Dept., 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from the specification may appear in the sample, in which case the specification shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

3.3 Material (see 6.5).

* 3.3.1 Uppers. The outer upper parts shall consist of quarters and tongue vamps made from cotton duck having a minimum weight of 12.29 ounces per square yard (417 g/m^2) conforming to type III of CCC-C-419, except that the cloth shall be bleached. The quarters and tongue vamps shall be lined with unbleached cotton army duck having a minimum weight of 6.50 ounces per square yard (220 g/m^2). Testing shall be as specified in 4.4.1.

3.3.2 Eyelet stays. The eyelet stays shall be cut from fabric having a minimum weight of 6.50 ounces per square yard (220 g/m^2) when tested as specified in 4.4.1. The eyelet stay shall be 3/4 to 1 inch (19 to 25 mm) wide.

* 3.3.3 Counter. The counter shall be an uncoated or resin coated cotton duck having a minimum weight of 7.0 ounces per square yard (237 g/m^2) or a resin coated osnaburg having a minimum weight of 6.0 ounces per square yard (203 g/m^2), combined with a rubber base curable stiffening compound. The overall thickness of the combined material shall be 0.080 to 0.090 inch (2.0 to 2.3 mm).

3.3.4 Backseam tape (inside). When used, the backseam tape shall be bias cut 3/8 to 1/2 inch (10 to 13 mm) in width from natural drill cotton.

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* 3.3.5 Back stay (outside). The back stay shall be made from not less than 3/4 inch (19 mm) and not more than 1 inch (25 mm) wide bleached cotton tape or from bleached woven cloth cut in strips with the edges folded to the center to form a tape not less than 3/4 inch (19 mm) and not more than 1 inch (25 mm) wide.

3.3.6 Innersoles. The bottom innersole shall be made of a rubber base stiffening compound and shall be not less than 0.050 inch (1.27 mm) in thickness.

3.3.7 Bias binding (for counter). The bias binding for the counter shall be sized cotton sheeting weighing approximately 2.8 ounces per square yard (95 g/m^2) when tested as specified in 4.4.1. As an alternate material, the bias counter binding may be a natural bias binding of good commercial quality. The binding for the counter shall be of sufficient width to permit a 9/32 $\pm$ 1/32 (7 $\pm$ 1 mm) fold around the top edge of the counter.

* 3.3.8 Thread polyester, polyester cotton or nylon. The thread shall be either polyester, polyester cotton or nylon. The polyester thread shall conform to type I or II, class 1, subclass A, size E of V-T-285, or cotton-covered polyester thread conforming to size 20, 3 ply of MIL-T-43548. The nylon thread shall conform to type I or II, size E of V-T-295. The color of the thread shall be white.

* 3.3.9 Laces. The laces shall conform to type II, class 1 of V-L-61. As an alternate, the laces may be cotton tubular laces having a minimum breaking strength of 100 pounds (45 kg). The length of the laces shall be 42 inches (107 cm) for shoe sizes 3 through 8-1/2 and 48 inches (122 cm) for shoe sizes 9 through 15. The color of the laces shall be white.

3.3.10 Eyelets.

3.3.10.1 Ventilating shank eyelets. The ventilating eyelets shall be size No. 1 or 2 aluminum eyelets.

3.3.10.2 Lacing eyelets and washers. The lacing eyelets shall be a telescopic number 404 or 606 aluminum eyelet with a number 202 or 646 aluminum washer.

3.3.11 Foxings. The foxings shall be made of natural rubber, synthetic rubber, or a mixture of natural and synthetic rubber.

3.3.11.1 First foxing. The first foxing shall have a minimum thickness of 0.035 inch (0.89 mm) and a width of 3/4 inch (19 mm) minimum. The foxing may be plain or engraved with the manufacturer's standard.

3.3.11.2 Second foxing. The second foxing shall be made of cotton osnaburg weighing a minimum of 4.0 ounces per square yard (136 g/m^2), frictioned on both

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sides, then coated with skin coat matching the first foxing. The combined thickness of the second foxing shall be a minimum of 0.045 inch (1.14 mm). The foxing may be plain or engraved with the manufacturer's standard.

3.3.12 Toe bumper strip and outside toe cap. The toe bumper strip and outside toe cap shall be made of natural rubber, synthetic rubber, or a mixture of natural and synthetic rubber. Rubber toe caps shall be adhered to the tongue-vamps on the outside. Rubber toe caps shall be a minimum of 0.035 inch (0.89 mm) thick. The toe bumper strip shall be a minimum of 0.085 inch (2.16 mm) thick. Shape and type of engraving shall be the manufacturer's standard.

* 3.3.13 Cushions. Each shoe shall have a full length cushion sole and cushion arch support made of sponge rubber. The compression set for the cushion arch support shall be not more than 35 percent and the compression deflection shall be between 16 and 20 psi (110 kPa and 138 kPa). The cushion sole shall be combined with cotton duck specified in 3.3.3.1 and shall have a minimum thickness of 0.125 inch (3.18 mm), and the cushion arch support extending from heel to slightly rearward of the ball line shall have a minimum thickness of 0.275 inch (6.98 mm) at the center of the arch. The cushion arch shall fit the arch of the last without distortion or bulging of the outersole in the arch area. The cushion sole and the cushion arch support may be combined with a suitable cement to form a single unit.

* 3.3.14 Outersole. The outersole shall be made of natural rubber, synthetic rubber, or a mixture of natural and synthetic rubber. The color of the outersoles shall be tan, brown, white, or natural. The outersole shall have a minimum thickness of 0.250 inch (6.35 mm) on top of engraving with the exception of the arch which shall have a minimum thickness of 0.175 inch (4.44 mm). The abrasive index shall be a minimum of 75 and the hardness shall be a minimum of 45 and a maximum of 65 when tested as specified in 4.4.1. The outersole shall be of the calendered type or premolded type and the manufacturer's standard design.

3.3.15 Adhesive. The cements and adhesives used in the various cementing operations shall consist of a suitable compound of natural rubber, synthetic rubber, thermoplastic type, or any suitable combinations thereof. The adhesive shall provide a permanent and even bond throughout the cemented areas of the combined parts and shall prepare and activate the components for vulcanization.

* 3.4 Design. The shoe shall be a low cut, loose lined, lace to toe oxford style with white duck uppers, white foxings and white toe cap. The tongue vamp shall be a bat-wing design with a seamless forepart. Each shoe shall have two ventilating eyelets on the inside shank. Size 3 through 8-1/2 shoes shall each have six pairs of eyelets and washers. Size 9 through 15 shoes shall each have seven pairs of eyelets and washers (see figure 1.).

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3.5 Patterns. Patterns shall be furnished by the contractor.

* 3.6 Lasts. The shoes furnished under this specification shall be made over the contractor's narrow and medium width lasts which shall be approved by the procuring activity concerned (see 6.2).

3.7 Stitches, seams, and stitchings. The stitch and seam types shall be as indicated in table I and shall conform to the requirements of FED-STD-751. The stitching shall be done with not less than 7 stitches and not more than 11 stitches to the inch (3 to 4 stitches/cm). Ends of all seams and stitchings, when not caught in other seams and stitchings or sealed by bottom construction, shall be backtacked not less than 1/2 inch (13 mm).

3.7.1 Repairs. Thread breaks and two or more consecutive skipped or run-off stitches shall be overstitched not less than 1/2 inch (13 mm) in each direction beyond the defective area.

3.8 Construction. The shoes shall be constructed in accordance with the operations listed in table I. The contractor is not required to follow the sequence of operations as listed, provided the finished shoes are identical with those produced by following the sequence as shown.

TABLE I. Manufacturing operations requirements

No.	Stitch type	Seam and stitching type
1. Join quarter linings and counters.		
a. Stitch heel edges of quarter lining together face to face. 3/16 to 1/4 inch (5 to 6 mm) gauge.	401	SSa-1
* Open seam and stitch down lap with one row of stitching on each side. Optional methods may be used to perform this operation. As an alternate, a one piece quarter lining may be used.	401	SSz-3
b. The counter with top edge binding tape shall be on the inside of the quarter lining (against the foot) and shall be stitched around the top edge of counter to the quarter lining with two rows of stitching 1/8 inch (3 mm) apart.	301	LSBj-2 (with tape)

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TABLE I. Manufacturing operations requirements (cont'd)

No.	Stitch type	Seam and stitching type
2. Join outside quarters at backseam closing.		
a. The backseam of the outside quarters shall be closed at the back, with zigzag stitching closely positioned to the edge of the quarters from top to bottom.	304	FSa-1 (with tape)
b. As an alternate, the zigzag stitching may be omitted and a backseam tape specified in 3.3.4 shall be used to cover the butted edges of the quarters. The tape shall be stitched on the inside of the quarters, using two rows of stitching with one row on each side of the backseam tape. Each row of stitches shall be 1/16 to 1/8 inch (2 to 3 mm) from edge of the tape.	401	LSp-2
3. Attach back stay to outside quarters.		
Back stay shall be stitched at the back of the quarters with four rows of stitching evenly spaced with the two outside rows closely positioned to the edges of the back stay.	301 or 401	-
4. Attach eyelet stay and eyelet row stitching.		
Attach eyelet stay on the inside of each quarter and stitch the bottom portion of the eyelet stay to the quarters with one row of stitching or two rows of stitching 1/8 inch (3 mm) apart.	301	SSaa-1 or 2
5. Join outside quarters and quarter lining.		
a. Superimpose the outside quarters and quarter lining face to face and stitch around top edge, using one row of stitching with a 1/4 inch (6 mm) lap.	301	SSa-1
b. Turn faces out and pound smoothly along the edge of stitching.		
c. The top edge of the outside quarters and quarter linings shall be stitched with one row of stitching or two rows of stitching 1/8 inch (3 mm) apart.	301	SSe-2 or 3

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TABLE I. Manufacturing operations requirements (cont'd)

No.	Stitch type	Seam and stitching type
6. Stitching tongue vamp lining to tongue vamp.		
The top and side edges of the combined tongue vamp lining to tongue vamp shall be over edge stitched to within 1/2 ± 1/8 inch (13 ± 3 mm) from the bottom of the tongue. The gage of stitching shall be 1/8 to 3/16 inch (3 to 5 mm).	505	SSa-1
7. Bottom stitch quarter lining and outside quarters.		
The quarter lining and outside quarters shall be pulled down and stitched to within 1/4 inch to 1/2 inch (6 to 13 mm) from bottom lasting edge with one row of stitching.	301 or 505	SSa-1 LFd-1
8. Attach combined quarters and vamp wing tongue.		
The vamp wing tongue and quarters shall be stitched with two rows of stitching 1/8 inch (3 mm) apart.	301	SSa-1
9. Attach eyelets and washers (lacing).		
Attach eyelet with washer through each side of quarters eyelet stay and quarter lining, equally spaced between the top and bottom and in the center of the eyelet row and top edge stitching. Size 3 through 8-1/2 shoes shall each have six pairs of eyelets and washers. Size 9 through 15 shoes shall each have seven pairs of eyelets and washers.		
10. Eyelet row opening.		
Optional methods may be used to perform this operation.		
11. Assembly.		
* a. The combined full length cushion sole with the assembled arch support shall be placed on the last. An upper of corresponding size shall be fitted over the last and shall be drawn and secured to the innersole.		

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TABLE I. Manufacturing operations requirements (cont'd)

No.	Stitch type	Seam stitching type
11. Assembly. (cont'd)		
Alternately, a socket lining of natural 6 ounces per square yard (203 g/m^2) drill rubber coated on one side may be sewed in and the shoe slip lasted in the conventional manner. The combined cushion sole is then securely cemented to the sock lining after vulcanization.		
b. Adhere the toe cap to the outside of the uppers.		
c. The first foxing or that portion of the lasted upper shall be cemented so that the foxing can adhere. The lower edge of the foxing shall be placed under the top edge of the outersole.		
d. Attach the outersole.		
e. The second foxing shall be even with or slightly above the outeredge of the outersole. The ends of the foxing shall be smoothly butted or overlapped up to a maximum of 3/8 inch (10 mm) at inner side in shank area or under the bumper. The second foxing shall be attached to the outersole and the first foxing. The width of the second foxing shall overlap the first foxing not less than 3/8 inch (10 mm).		
f. The bumper foxing shall be attached around the toe, extending beyond the first eyelet on each side.		
12. Vulcanization.		
The shoes shall be vulcanized under pressure to ensure a completely sealed unit.		
13. Attach ventilating eyelets.		
Attach two ventilating eyelets on the inside shank of each shoe located just above the first foxing on the finished shoe and spaced 1 ± 1/4 inch (25 ± 6 mm) from center eyelet to center eyelet.		

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TABLE I. Manufacturing operations requirements (cont'd)

No.	Stitch type	Seam stitching type
14. Finish.		
The shoes shall have a natural heater finish.		
15. Marking quarters.		
The contractor's name, contract number, and correct size of the shoe shall be legibly stamped with black ink. Stamping shall be a minimum of 18 points (approximately 1/4 inch) (6 mm). Marking shall be applied on the inside of each shoe on the quarter lining or counter.		
16. Laces.		
The shoes shall be mated. A lace shall be inserted through an eyelet of each shoe and the two laces shall be tied firmly together.		
17. Thread ends.		
All thread ends shall be trimmed throughout shoe.		

3.8.1 Replacement of defective components. During the cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.

3.9 Workmanship. The finished shoes shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform

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any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2), the first article submitted in accordance with 3.2 shall be visually inspected as specified in 4.4.3 for compliance with design, construction, and workmanship requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, testing shall be performed on components and materials listed in table II for the characteristics shown. The sample unit for testing of fabric shall be 1/4 yard (230 mm) full width. The sample unit for testing of rubber compounds shall be taken at random from one of the rubber batches used in a day's production and shall consist of a slab 3 by 6 by 0.025 inches (76 mm by 152 mm by 0.64 mm), vulcanized in the same manner as the shoe. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size for the fabrics shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. Component tests

Component	Characteristic	Requirement paragraph	Test method
Cotton duck quarter linings and tongue vamp lining)	Material identification	3.3.1	1200 1/
	Weight	3.3.1	5041 1/
Eyelet stay	Weight	3.3.2	5041 1/
Counter fabric	Material identification	3.3.3	2/
	Weight	3.3.3	2/
Innersole	Thickness	3.3.6	2/
Binding fabric	Material identification	3.3.7	1200 1/
	Weight	3.3.7	5041 1/
Laces (alternate material)	Material identification	3.3.9	2/
	Breaking strength	3.3.9	2/
Cushion arch support	Compression set	3.3.13	D 1056-73 3/ 4/
	Compression deflection	3.3.13	D 1667-76 3/
Outersole compound	Abrasive index	3.3.14	D 1630-61 3/
	Hardness	3.3.14	3021 5/
Adhesive	Material identification	3.3.15	2/

1/ Refers to FED-STD-191.

2/ A certificate of compliance shall be submitted and will be acceptable for the stated requirements.

3/ Refers to ASTM Standard.

4/ Compression set will be measured under constant deflection at 50 percent of original thickness for 48 hours at 70° ± 2°C (158° ± 3.6°F). Reading taken 30 minutes after release and rest at room temperature.

5/ Refers to FED-STD-601.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are

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accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance.

4.4.3 Examination of the end item. The defects found during examination shall be classified in accordance with the list below. The sample unit shall be one completely fabricated shoe and the selection shall be by pairs. The lot size shall be expressed in units of one shoe. For pairing examination, the sample units constituting one pair shall be examined together. Defects for pairing shall be classified as a single defect. The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 major and 6.5 total defects per hundred units.

Classification of defects

Examine	Defect	Classification Major	Minor
Pairing	Not properly mated, i.e., not right and left of same size.	X	
	Variation between shoes:		
	-seriously affecting appearance.	X	
	-affecting appearance but not seriously.		X
Cleanness	Any nonremovable spot or stain other than that appearing on any tread surface.	X	
	Any spot or stain removable with cleaning agent.		X
Color	Not specified color.	X	
	Color not uniform.		X
Design and size	Design not as specified.	X	
Construction and workmanship	Any cut, tear, hole, mend, or weak place.	X	
	Any component or assembly omitted or misplaced, operation omitted or not properly performed (unless otherwise classified herein) i.e., rubber toe cap missing or out of alignment; tongue misplaced to an extent that will cause discomfort to wearer:		
	-seriously affecting serviceability or appearance.	X	
	-affecting serviceability or appearance but not seriously.		X

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Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams and stitching	Not specified seam or stitch type.	X	
	Any row of stitching missing.	X	
	Ends of seams or stitchings, when not caught in other seams or stitchings, not backtacked or backtacked less than 1/2 inch (13 mm).		X
	Raw edge, which should be concealed, showing.		X
	Irregular or uneven stitching affecting appearance.		X
	More than 13 stitches per inch.	X	
	More than 11 but not more than 13 stitches per inch.		X
	Less than 7 stitches per inch.	X	
	Loose stitch tension resulting in a loosely secured seam.	X	
	Tight stitch tension resulting in puckering of fabric or assembly.	X	
	NOTE: A plus tolerance of 3 stitches per inch will be allowed when stitching around top of quarter eyestay curve.		
	Gage of stitching or stitching margin not as specified.		X
	Open seam not repaired. 1/	X	
	Repair of open seam not as specified. 1/		X
	Thread ends not trimmed throughout shoe.		X
Rubber components	Not securely bonded where required.	X	
	Porous, imbedded foreign matter, void, or blister:		
	-seriously affecting serviceability or appearance.	X	
	-affecting serviceability or appearance but not seriously.		X

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Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Eyelets, lacing	Any eyelet not securely clinched, i.e., can be turned by hand.	X	
	Eyelets not properly spaced within the row or misalignment between the rows to an extent interfering with proper lacing:		
	-seriously affecting serviceability or appearance.	X	
	-affecting serviceability or appearance but not seriously.		X
	Number of eyelets not as specified but each row has the same number.		X
Ventilating eyelets and washers	Any ventilating eyelet or washer missing.	X	
	Not securely clinched, or not inserted as specified.		X
Marking	Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible, or size not as specified.		X

1/ A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off occur. occur. On double stitched seams, a seam is considered open when either one or both sides of the seam are open.

4.4.4 Packaging inspection. An examination shall be made to determine that the preservation-packaging, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Examination for the closure defects listed below shall be made on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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Examine	Defect
Marking (exterior and unit packs)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container. Polyethylene bag heat sealed seam open, noncontinuous, or crooked. Incorrectly fabricated bag. Tissue paper improperly positioned or missing.
Content	Number of pairs of shoes per container is more or less than required. Size indicated on shoes not the same as specified on packages or shipping containers. 1/

1/ For this defect, one package from each container in the sample shall be examined.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A. Each pair of shoes shall be placed heel to tow or toe to toe with the uppers facing each other, than inserted in a snug-fitting clear polyethylene film bag of 0.0015 inch (0.038 mm) thickness (± 25 percent tolerance). The polyethylene bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag. The bag may be fabricated from polyethylene film tubing or sheeting. One corner of the bag may have a 1/4 inch (6 mm) diameter hole to permit expulsion of excess air from within the bag. Closure of the polyethylene bag shall be accomplished by means of a mechanical tie (paper or plastic covered soft steel wire, aluminum band, etc.). Alternately, the closure of the polyethylene bag may be accomplished by means of a heat seal.

5.1.2 Level C. Shoes shall be preserved-packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A packing. Twelve pairs of shoes of one size and width only, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A bagged shoes shall be packed flat, in layers, with the length of the packages parallel to the end panels of the container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

* 5.2.2 Level B packing. Twelve pairs of shoes of one size and width only, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 200 of PPP-B-636. Level A bagged shoes shall be packed flat, in layers, with the length of the packages parallel to the end panels of the container. Each shipping container shall be closed in accordance with method II as specified in PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container may be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.3 Level C packing. Shoes, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with Uniform Freight Classification, or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract, unit packs and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Polyethylene bagged shoes. Polyethylene bagged shoes shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag, so as to permit ready identification.

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5.3.2 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches (130 by 100 mm) with the words "MIXED SIZES" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity of pairs and sizes contained therein.

6. NOTES

* 6.1 Intended use. The shoes are intended for use by personnel of the Department of Defense when involved in athletic activities.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Sizes and widths required (see 1.2).
(c) When a first article is required (see 3.2, 4.3, and 6.4).
(d) One pair of contractor's lasts in narrow and medium widths (any size) shall be submitted to the contracting officer for approval (see 3.6).
(e) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).
(f) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Sample. For access to samples, address the procuring activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one pair of shoes. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

* 6.6 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

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* 6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Air Force - 45, 99
DLA - CT

Preparing activity:

Army - GL

Project No. 8430-0262

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FIGURE 1 SHOES, GYMNASIUM, HEAVY DUTY

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-S-43961A Shoes, Gymnasium, Heavy Duty

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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